

Experimental Verification of Automated Loss Simulation of Magnetic Components

Losses predicted by an automated magnetic simulation are checked by measuring the heat they produce

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THE PRINCIPLE

$$P_{\text{loss}} = C_{\text{Th}} \cdot dT/dt$$

Every watt lost in the core and winding of a magnetic component ends up as heat. Measuring how fast a known heat capacity C_{Th} warms up gives the real loss — independent of the simulation.

C_{Th} : total heat capacity of the stirred liquid, the enclosure and the device under test.



1 • Magnetic simulation

PLECS, Infineon SPICE models and 2D FEM in Ansys Maxwell predict the core and winding losses of inductors and transformers.



2 • Thermal measurement

The inductor runs in its converter inside a stirred, near-adiabatic liquid calorimeter; its temperature slope gives the real loss.



3 • Comparison

Buck converter, RM14LP, 400 kHz: litz within 1.5 %, solid wire measured 8–11 % above simulation.

Magnetic side: automated loss simulation

Python-based workflow: PLECS waveforms → FFT → Ansys Maxwell 2D losses of the magnetic component

1

Circuit model **PLECS**

The converter (example: buck converter) is drawn in PLECS; the waveforms of the magnetic component are extracted.

2

Switch losses **Infineon SPICE**

Switch losses are simulated with Infineon device models: ≈ 6.8 W per switch.

3

FFT of the current **Python**

An FFT of the inductor current is done; the 10 most dominant frequency components are kept.

4

Magnetic FEM **Ansys Maxwell 2D**

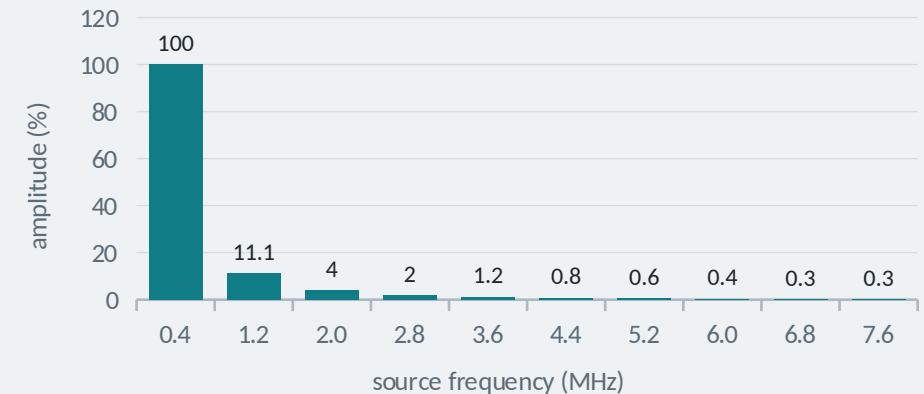
They become parallel sinusoidal sources. Winding layer stack and material (litz or solid), core size & material, insulation thicknesses and air gaps are set.

5

Loss result **core + winding**

Total inductor loss: 6.75 W with solid wire, 6.70 W with litz wire — the quantity that is then measured as heat.

FFT step: the 10 sinusoidal sources



Own illustration (ideal triangular ripple at 400 kHz, $D = 0.5$): amplitudes fall with $1/n^2$ (100, 11, 4, 2 ... 0.3 %), so these 10 sources rebuild the current within 2 % of its peak — “good results and acceptable simulation time” (poster).

CASE STUDY

Buck converter • RM14LP core • 400 kHz • ≈ 880 W

≈ 6.8 W

loss per switch

6.75 W

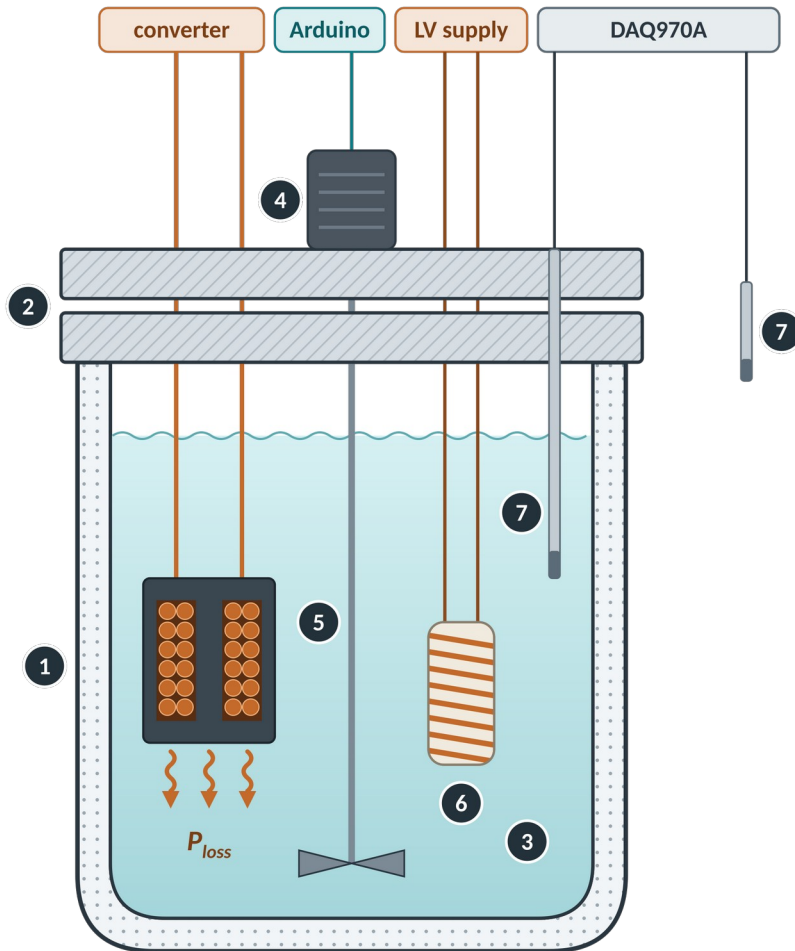
inductor, solid wire

6.70 W

inductor, litz wire

Thermal side: measuring the losses as heat

Near-adiabatic calorimeter: the inductor's losses heat a stirred liquid of known heat capacity



$$P_{\text{loss}} = C_{\text{Th}} \cdot dT/dt \approx C_{\text{Th}} \cdot \Delta T / \Delta t$$

C_{Th} : total heat capacity inside the adiabatic enclosure; ΔT : its temperature change during the time Δt in which the power is dissipated. Valid because almost no heat leaves the Dewar.

Set-up

- 1 Dewar jar (vacuum-insulated)
- 2 Custom double lid
- 3 Insulating liquid, stirred
- 4 Stepper motor + Arduino (stirrer)
- 5 DUT: inductor, fully submerged
- 6 LV supply + wire-wound resistor
- 7 PT100 (fluid, ambient) → DAQ970A

Why this design?



Losses become heat

Core and winding losses are captured together, independent of waveform and frequency [3, 6].



Near-adiabatic

Unlike other calorimetric methods, no mass-flow measurement is necessary [4, 5].



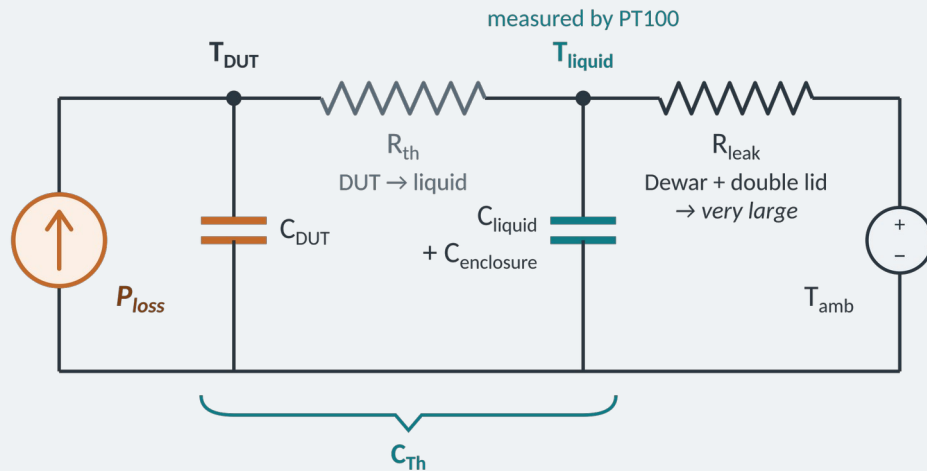
Stirred liquid

Good electrical insulation and high heat capacity at reasonable cost; stirring heats it uniformly.

Thermal model and calibration

Known heater powers turn the temperature slope into watts — under the same thermal conditions as the DUT

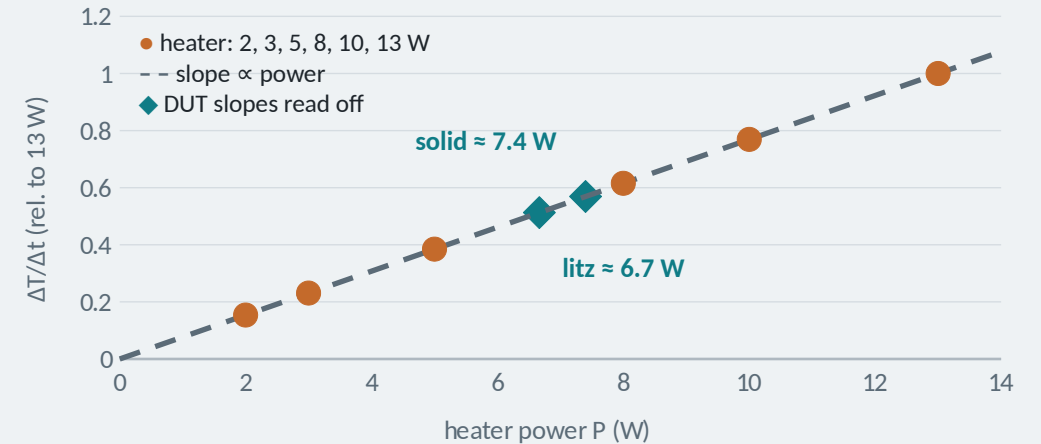
Lumped thermal model of the calorimeter



$$C_{Th} \cdot dT/dt = P_{loss} - (T - T_{amb})/R_{leak} \rightarrow P_{loss} \approx C_{Th} \cdot \Delta T/\Delta t \text{ for large } R_{leak}$$

- C_{Th} includes the DUT → it is fully submerged also during calibration
- R_{th} delays the response → only the most linear part of $\Delta T(t)$ is used
- Leakage is small, not zero → every run starts at 35 °C
- One liquid temperature → continuous stirring keeps the liquid uniform

Calibration: slope $\propto$ power

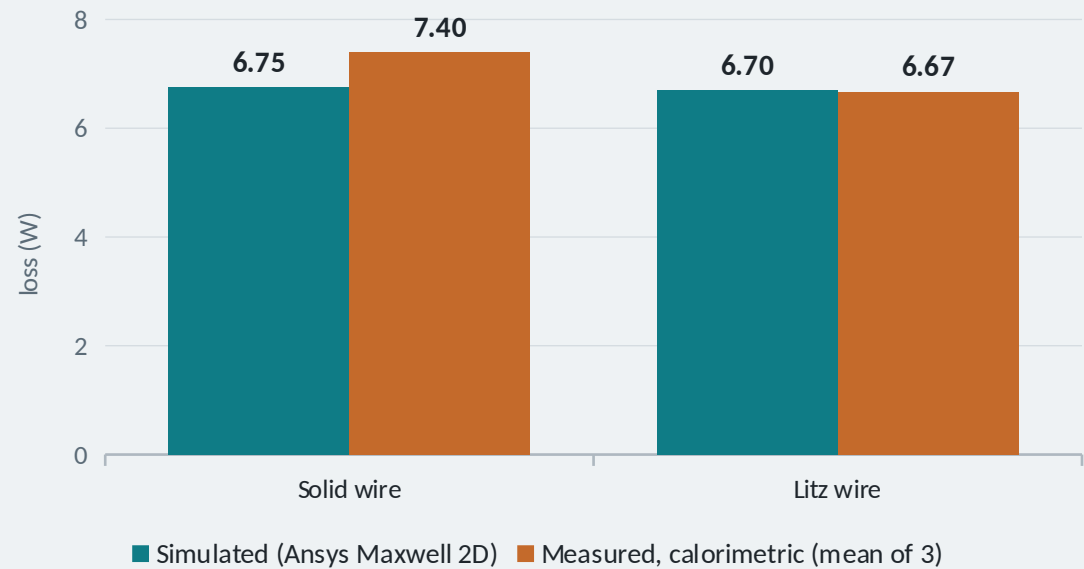


- 1 Calibrate:** LV supply + wire-wound resistor at known power (4-wire measurement)
- 2 Fit:** most linear region of each $\Delta T(t)$ record → one $\Delta T/\Delta t$ slope per power
- 3 Measure:** DUT operated as in normal operation → its $\Delta T/\Delta t$ slope
- 4 Interpolate:** loss read between the calibrated slopes

Results: simulated vs. thermally measured losses

Litz-wire inductor agrees within 1.5 %; the solid-wire inductor is measured 8–11 % above simulation

Inductor loss — RM14LP, 400 kHz, ≈ 880 W



Wire	Run 1	Run 2	Run 3	Mean	Simulated	Meas. vs. sim.
Solid	7.3	7.4	7.5	7.40	6.75	+9.6 %
Litz	6.7	6.7	6.6	6.67	6.70	-0.5 %

Six calorimetric runs from the poster, values in W. Deviation = (measured mean – simulated) / simulated.

LITZ WIRE

-0.5 %

measured vs. simulated (mean)
per run: -1.5 % ... 0.0 %

SOLID WIRE

+9.6 %

measured vs. simulated (mean)
per run: +8.1 % ... +11.1 %



Accurate loss prediction

The workflow predicts the losses of the litz-wound inductor very close to the thermally measured values.



Solid-wire effect under-represented

Measured: solid is 0.73 W above litz; simulated: only 0.05 W. Why this occurs is left for future analysis [10].



Good approximation for inductor design

Measured losses are within ~11 % of the simulated losses; the calorimeter resolves 0.1 W differences between runs.

References

The poster, the reference it cites, and related work on thermal loss measurement and magnetic loss modelling

The poster and the reference it cites

- [1] P. Skoff, M. Stoiber, J. Reynvaan, and W. Konrad, "Experimental verification of automated loss-simulation of magnetic components," poster, *Power Electronics for Energy Transition Symposium*, Vienna, Austria, Sep. 29–30, 2026.
- [2] J. Reynvaan, M. Pajnić, and J. Krenn, "Evaluating fluid based transient calorimetric method for measurement of the ferrite core losses," in *Proc. ICPE 2023-ECCE Asia*, Jeju, Korea, 2023, pp. 3308–3313, doi: 10.23919/ICPE2023-ECCEAsia54778.2023.10213808.

Thermal (calorimetric) loss measurement – similar concept

- [3] P. Papamanolis, T. Guillod, F. Krismer, and J. W. Kolar, "Transient calorimetric measurement of ferrite core losses up to 50 MHz," *IEEE Trans. Power Electron.*, vol. 36, no. 3, pp. 2548–2563, 2021.
- [4] D. Christen, U. Badstuebner, J. Biela, and J. W. Kolar, "Calorimetric power loss measurement for highly efficient converters," in *Proc. IPEC 2010 (ECCE Asia)*, Sapporo, Japan, 2010, pp. 1438–1445.
- [5] S. Weier, M. A. Shafi, and R. McMahon, "Precision calorimetry for the accurate measurement of losses in power electronic devices," *IEEE Trans. Ind. Appl.*, vol. 46, no. 1, pp. 278–284, 2010.
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Magnetic loss and thermal modelling / simulation – similar concept

- [9] P. L. Dowell, "Effects of eddy currents in transformer windings," *Proc. IEE*, vol. 113, no. 8, pp. 1387–1394, 1966.
- [10] R. P. Wojda and M. K. Kazimierczuk, "Winding resistance and power loss of inductors with litz and solid-round wires," *IEEE Trans. Ind. Appl.*, vol. 54, no. 4, pp. 3548–3557, 2018.
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- [12] P. A. Kyaw, M. Delhommais, J. Qiu, C. R. Sullivan, J.-L. Schanen, and C. Rigaud, "Thermal modeling of inductor and transformer windings including litz wire," *IEEE Trans. Power Electron.*, vol. 35, no. 1, pp. 867–881, 2020.
- [13] N. Förster et al., "An open-source FEM magnetic toolbox for calculating electric and thermal behavior of power electronic magnetic components," in *Proc. EPE'22 ECCE Europe*, Hanover, Germany, 2022.
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- [15] T. Guillod, P. Papamanolis, and J. W. Kolar, "Artificial neural network (ANN) based fast and accurate inductor modeling and design," *IEEE Open J. Power Electron.*, vol. 1, pp. 284–299, 2020.

HOW THE REFERENCES CONNECT TO THE SLIDES

Thermal measurement (slides 3–4): [2] same group's fluid calorimeter, [3] transient calorimetry, [4]–[8] other loss-measurement methods

Magnetic simulation (slide 2): [9]–[11] winding and core loss models, [13]–[15] automated FEM workflows

Litz vs. solid wire (slide 5): [10] winding loss, [12] winding thermal model